

MANISH SHAH

Physics | Green Hydrogen (Water Electrolysis), Nanomaterials & Graphene Gas Sensors, Renewable Energy Systems
Lalitpur, Nepal | Nepalese citizen | +977 9861360228 | manish2020shah@gmail.com | Portfolio: www.shahmanish.com.np

ACADEMIC PROFILE

- Physics graduate (B.Sc., Physics and Mathematics) from Patan Multiple Campus, Institute of Science and Technology (IoST), Tribhuvan University, Nepal (60.15%).
- Hands-on experimental researcher working on green hydrogen production by water electrolysis, green-synthesised copper nanomaterials and graphene-based gas sensors, with a focus on low-cost, locally buildable energy and sensing systems for Nepal.

RESEARCH INTERESTS

- Green hydrogen and water electrolysis; electrode design and electrocatalysis (HER/OER); green synthesis of metal nanoparticles; graphene and 2D materials for gas sensing; renewable energy integration and hydrogen utilisation; experimental physics and instrumentation.

EDUCATION

- B.Sc. in Physics (majors: Physics, Mathematics, Statistics), Patan Multiple Campus, IoST, Tribhuvan University, Patan Dhoka, Lalitpur, Nepal. 60.15%.
- Higher Secondary (+2, Science), Shikshadeep Higher Secondary School, Biratnagar, Morang, Nepal (NEB). Subjects: Physics, Chemistry, Mathematics, Computer Science. GPA 3.06.
- School (SLC/SEE), Lahan Merry Children Academy, Lahan, Siraha, Nepal. Subjects: Mathematics, Science, English, Optional Mathematics, Computer Science. GPA 3.30.

RESEARCH EXPERIENCE

- B.Sc. project thesis (PRO-406), Department of Physics, Patan Multiple Campus, TU (submitted August 2024): "Green Hydrogen Production via Electrolysis with Two–Three Electrode Model." Supervisor: Asst. Prof. Santosh Kumar Das; co-supervisor: Mr. Saddam Husain Dhobi (Central Department of Physics, TU).
 - Designed and built two-electrode and three-electrode (working, counter, reference) alkaline electrolysis cells using 99%-purity graphite electrodes (15 mm × 75 mm) in NaCl electrolyte.
 - Built the DC supply chain: step-down transformer (500 VA, 0–30 V AC), bridge rectifier and variac for a 5–30 V DC range; measured current and voltage; collected H₂ by downward displacement.
 - Ran a parametric study of electrolyte concentration (0–5 g NaCl), inter-electrode distance (1, 2, 3 mm), applied voltage/power and electrode alignment (parallel vs inclined).
 - Found that the three-electrode cell reached ~200 mL of H₂ against ~135 mL for the two-electrode cell (5 g NaCl, 1 mm gap); yield rose with concentration and power, fell with electrode gap, and was higher for parallel alignment.
 - Validated trends against published graphite-electrode electrolysis data (Yuvaraj & Santhanaraj, 2014) and wrote the literature review on AWE, PEM, AEM and solid-oxide electrolyzers.
- Researcher, Innovation Ghar Nepal, Lalitpur (February 2023 – present).
 - Handle and analyse experimental and research data; prepare research proposals, articles and manuscripts for publication.
- Green Synthesis of Copper Nanoparticles from Waste Materials.
 - Prepared copper nanoparticles by a green synthesis route using waste materials and studied their applications.
- Synthesis of Graphene and Fabrication of a Graphene Gas Sensor.
 - Designed and modelled the fabrication route (CVD growth on copper foil, transfer, patterned electrodes) and costed and scheduled the experiment.
 - Carried out sensitivity analysis of graphene sensor parameters; basic Arduino programming for read-out.
- Further project work: hydrogen purification and power generation; thermal evaluation of power generation and of an evacuated solar-tube receiver; renewable-energy integration.
- Poster presentation and project demonstration, 6th National Science Day, Nepal Academy of Science and Technology (NAST), Khumaltar, Lalitpur.

PROFESSIONAL EXPERIENCE

- Data Entry, Central Bureau of Statistics, Nepal (29 Nov 2021 – 14 Feb 2022): transferred paper records to the computer database, maintained data accuracy and followed data protocols.
- Teacher, Peace Garden English Boarding School, Khokana (16 Mar 2019 – 14 Jun 2021): prepared and delivered lessons using varied teaching methods; prepared assignments.

TECHNICAL SKILLS

- Laboratory: electrolysis cell design and assembly (two- and three-electrode), DC power-supply set-up, gas collection and volumetric measurement, nanoparticle green synthesis.
- Programming and electronics: C, QBASIC, basic Python; Arduino programming and data-logger construction.
- Design and modelling: AutoCAD (working drawings), CATIA (turbine blade and hub), SOLIDWORKS (biogas system, gears, machine parts).
- Documentation and media: MS Word, Excel, PowerPoint (proposals, reports, conference papers); Adobe Photoshop; DaVinci Resolve.

TRAINING

- Training on Deep Learning in Physics (15 days).
- Workshop on Python using a Supercomputer (5 days).
- Training on Research Methodology and Research Paper Writing/Publication Process, Centre for Energy Studies (CES) and University Grants Commission (UGC), Nepal (2 days).
- Diploma in Computer and Basics (6 months).

LANGUAGES

- Nepali, English, Maithili and Hindi (fluent, written and spoken).

REFERENCES

- Mr. Saddam Husain Dhobi, Central Department of Physics, Tribhuvan University. Mobile: +977 9808100285.
- Er. Shailendra Shah, National Vigilance Centre, Singhadurbar, Kathmandu. Mobile: +977 9845457100.